



Lecture # 9

Project Scope Management

ENGINEERING PROJECT MANAGEMENT

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Learning Outcomes

- ▶ Describe an overall framework for project scope management
 - ▶ Plan Scope Management
 - ▶ Collect Requirements
 - ▶ Define Scope
 - ▶ Create work breakdown structure - WBS
 - ▶ Validate Scope
 - ▶ Control Scope

Project Scope Management

- A Knowledge Area is an identified area of project management defined by its **knowledge requirements** and described in terms of its component processes, practices, inputs, outputs, tools, and techniques.

Source: PMI, 2017, "A guide to the project management body of knowledge (PMBOK guide)," **Sixth edition**, Newtown Square, PA: Project Management Institute, Inc.

What is Project Scope Management?

- ▶ **Scope** refers to *all* the work involved in creating
 - ▶ The products of the project and
 - ▶ The processes [**all project work**] used to create them (project)

Project Scope Management Processes

- ▶ **Plan Scope Management**—The process of creating a **scope management plan** that documents how the project and product scope will be defined, validated, and controlled.
- ▶ **Collect Requirements**—The process of determining, documenting, and managing **stakeholder** needs and requirements to meet project objectives.
- ▶ **Define Scope**—The process of developing a **detailed description** of the project and product.

Project Scope Management Processes

- ▶ **Create WBS**—The process of subdividing project deliverables and project work into smaller, more manageable components.
- ▶ **Validate Scope**—The process of formalizing acceptance of the completed project deliverables.
- ▶ **Control Scope**—The process of monitoring the status of the project and product scope and managing changes to the scope baseline

Table 1-4. Project Management Process Group and Knowledge Area Mapping

Knowledge Areas	Project Management Process Groups				
	Initiating Process Group	Planning Process Group	Executing Process Group	Monitoring and Controlling Process Group	Closing Process Group
4. Project Integration Management	4.1 Develop Project Charter	4.2 Develop Project Management Plan	4.3 Direct and Manage Project Work 4.4 Manage Project Knowledge	4.5 Monitor and Control Project Work 4.6 Perform Integrated Change Control	4.7 Close Project or Phase
5. Project Scope Management		5.1 Plan Scope Management 5.2 Collect Requirements 5.3 Define Scope 5.4 Create WBS		5.5 Validate Scope 5.6 Control Scope	
6. Project Schedule Management		6.1 Plan Schedule Management 6.2 Define Activities 6.3 Sequence Activities 6.4 Estimate Activity Durations 6.5 Develop Schedule		6.6 Control Schedule	
7. Project Cost Management		7.1 Plan Cost Management 7.2 Estimate Costs 7.3 Determine Budget		7.4 Control Costs	
8. Project Quality Management		8.1 Plan Quality Management	8.2 Manage Quality	8.3 Control Quality	

Plan Scope Management

- ▶ The key benefit of this process is that it provides guidance and direction on how scope will be managed throughout the project

Plan Scope Management

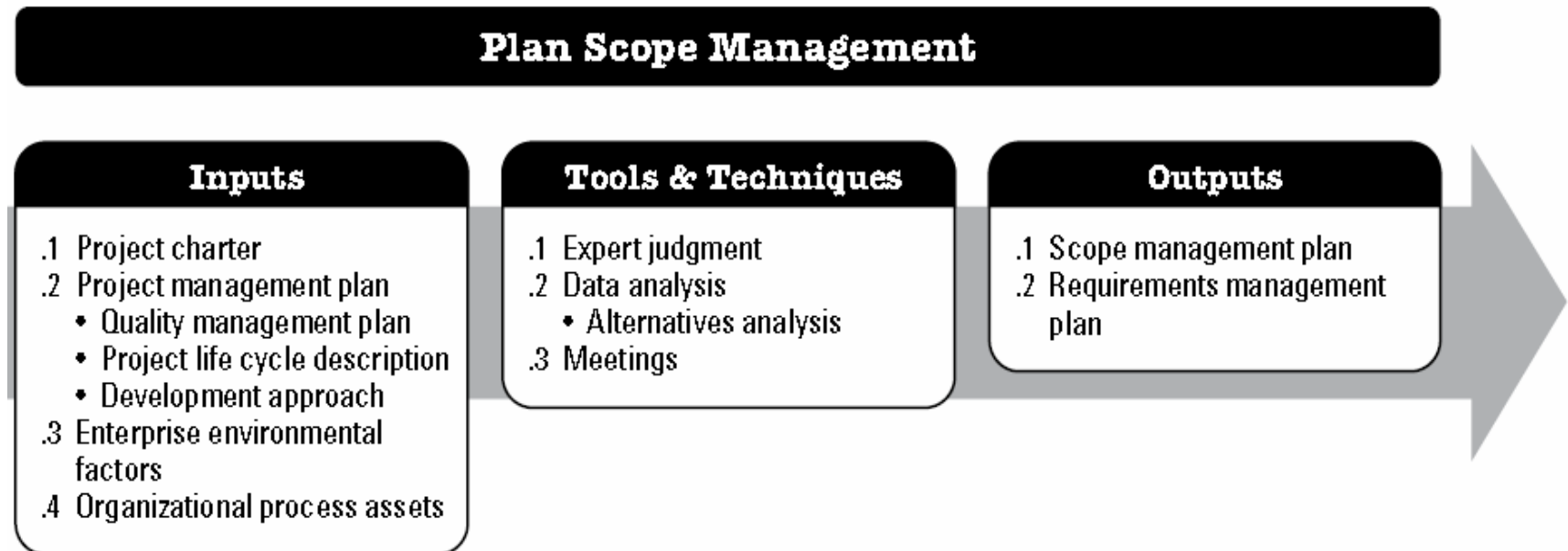


Figure 5-2. Plan Scope Management: Inputs, Tools & Techniques, and Outputs

Plan Scope Management

- ▶ The components of a **scope management plan** include:
 - ▶ **Process** for preparing a project scope statement
 - ▶ **Process** that enables the creation of the WBS from the detailed project scope statement
 - ▶ **Process** that establishes how the scope baseline will be approved and maintained and
 - ▶ **Process** that specifies how formal acceptance of the completed project deliverables will be obtained.

Collect Requirements

- ▶ The key benefit of this process is that it provides the basis for defining the product scope and project scope.

Collect Requirements

Collect Requirements

Inputs

- .1 Project charter
- .2 Project management plan
 - Scope management plan
 - Requirements management plan
 - Stakeholder engagement plan
- .3 Project documents
 - Assumption log
 - Lessons learned register
 - Stakeholder register
- .4 Business documents
 - Business case
- .5 Agreements
- .6 Enterprise environmental factors
- .7 Organizational process assets

Tools & Techniques

- .1 Expert judgment
- .2 Data gathering
 - Brainstorming
 - Interviews
 - Focus groups
 - Questionnaires and surveys
 - Benchmarking
- .3 Data analysis
 - Document analysis
- .4 Decision making
 - Voting
 - Multicriteria decision analysis
- .5 Data representation
 - Affinity diagrams
 - Mind mapping
- .6 Interpersonal and team skills
 - Nominal group technique
 - Observation/conversation
 - Facilitation
- .7 Context diagram
- .8 Prototypes

Outputs

- .1 Requirements documentation
- .2 Requirements traceability matrix

Figure 5-4. Collect Requirements: Inputs, Tools & Techniques, and Outputs

Affinity diagram: Organizes a large number of ideas into their natural relationships.

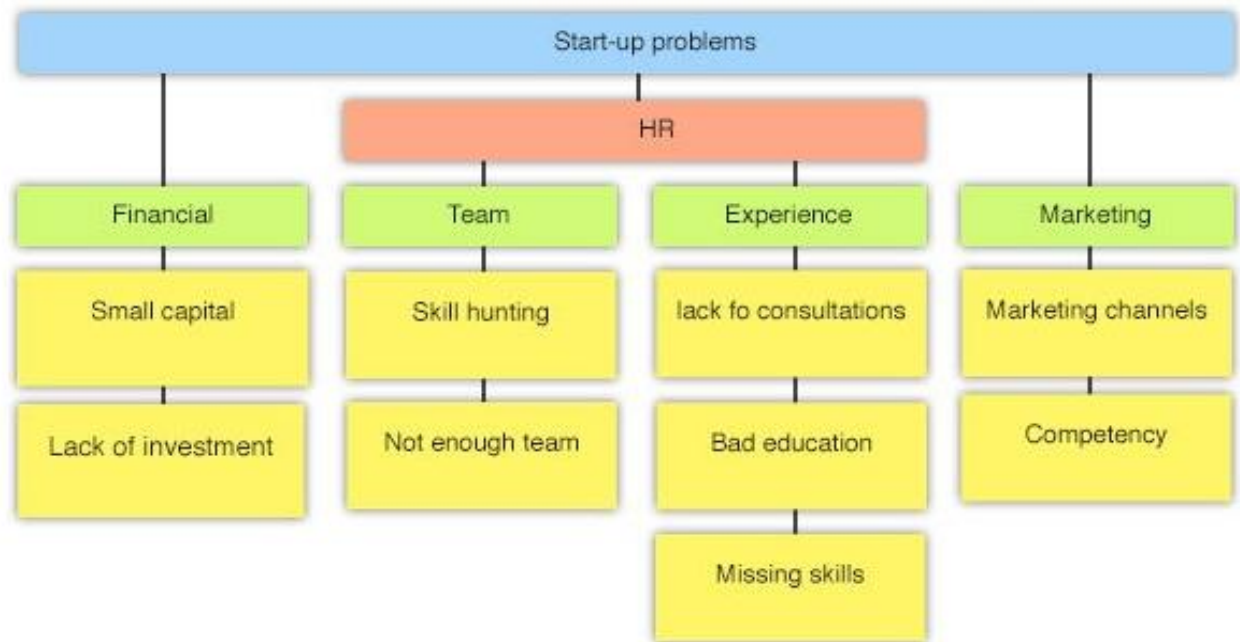
Interrelationship diagram: Shows cause-and-effect relationships and helps analyze the natural links between different aspects of a complex situation.

Tree diagram: Breaks down broad categories into finer and finer levels of detail, helping to move step-by-step thinking from generalities to specifics.

Matrix diagram: Shows the relationship between two, three, or four groups of information and can give information about the relationship, such as its strength, the roles played by various individuals, or measurements.

Affinity Diagram

Start-up Problems - Affinity Diagram



Requirements Documentation

Requirements Matrix for Project Name

↵

Prepared by:

Date: ↵

↵

Requirement No. ↵	Name ↵	Category ↵	Source (for Who?) ↵	Status ↵
↵	↵	↵	↵	↵
↵	↵	↵	↵	↵
↵	↵	↵	↵	↵
↵	↵	↵	↵	↵
↵	↵	↵	↵	↵
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Methods for Evaluating Requirements

- There are three ways of evaluating votes in decision-making:
 - **Unanimity** (agreement by all involved)
 - **Majority** (> 50% of agreements)
 - **Plurality** (the most agreed option)

Requirements Traceability Matrix (RTM)

Requirements Traceability Matrix								
Project Name:								
Cost Center:								
Project Description:								
ID	Associate ID	Requirements Description	Business Needs, Opportunities, Goals, Objectives	Project Objectives	WBS Deliverables	Product Design	Product Development	Test Cases
001	1.0							
	1.1							
	1.2							
	1.2.1							
002	2.0							
	2.1							
	2.1.1							
003	3.0							
	3.1							
	3.2							
004	4.0							
005	5.0							

Figure 5-7. Example of a Requirements Traceability Matrix

Define Scope

- ▶ The key benefit of this process is that it provides the basis for defining the product scope and project scope

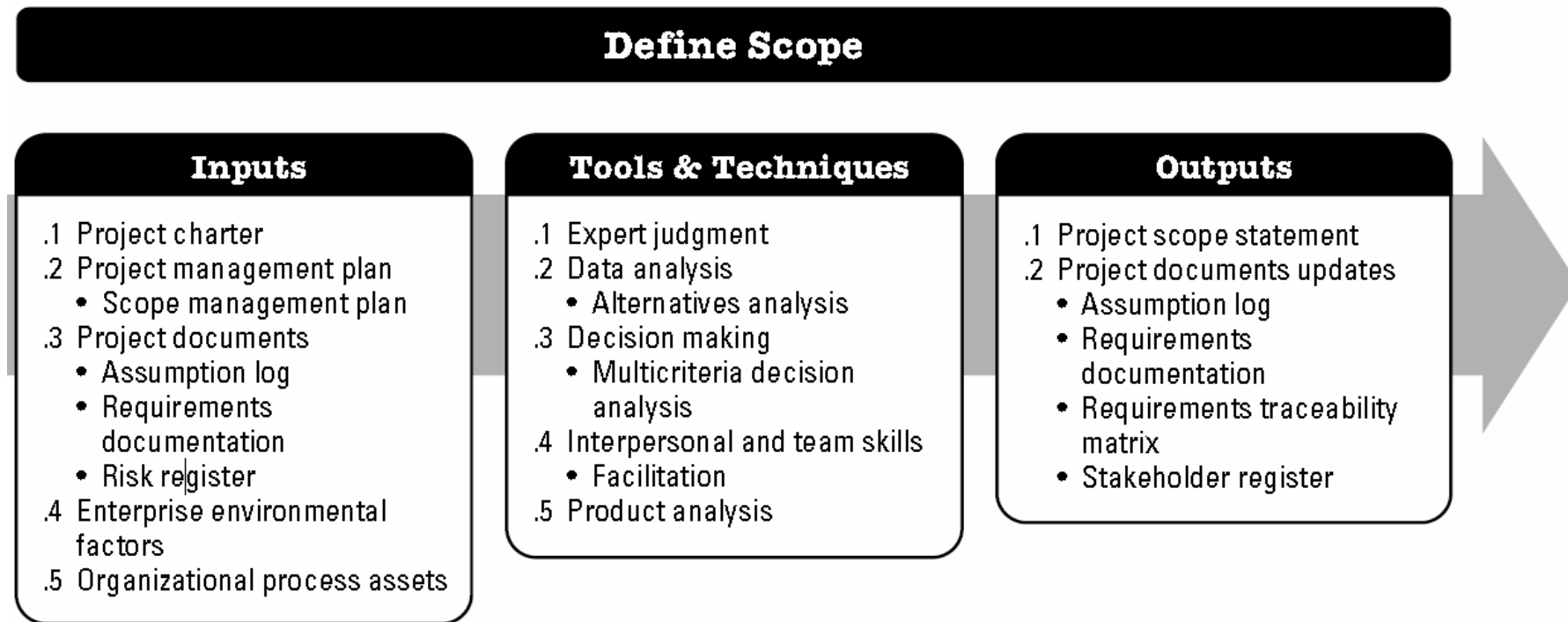


Figure 5-8. Define Scope: Inputs, Tools & Techniques, and Outputs

Table 5-1. Elements of the Project Charter and Project Scope Statement

Project Charter

Project purpose

Measurable project objectives and related success criteria

High-level requirements

High-level project description, boundaries, and key deliverables

Overall project risk

Summary milestone schedule

Preapproved financial resources

Key stakeholder list

Project approval requirements (i.e., what constitutes success, who decides the project is successful, who signs off on the project)

Project exit criteria (i.e., what are the conditions to be met in order to close or to cancel the project or phase)

Assigned project manager, responsibility, and authority level

Name and authority of the sponsor or other person(s) authorizing the project charter

Project Scope Statement

Project scope description (progressively elaborated)

Project deliverables

Acceptance criteria

Project exclusions

Create WBS

- ▶ The key benefit of this process is that it provides a framework of what has to be delivered.
- ▶ The WBS is
 - ▶ a hierarchical decomposition of the total scope of work to be carried out by the project team
 - ▶ to accomplish the project objectives and create the required deliverables.
 - ▶ defines the total scope of the project and
 - ▶ represents the work specified in the current approved project scope statement

Create WBS

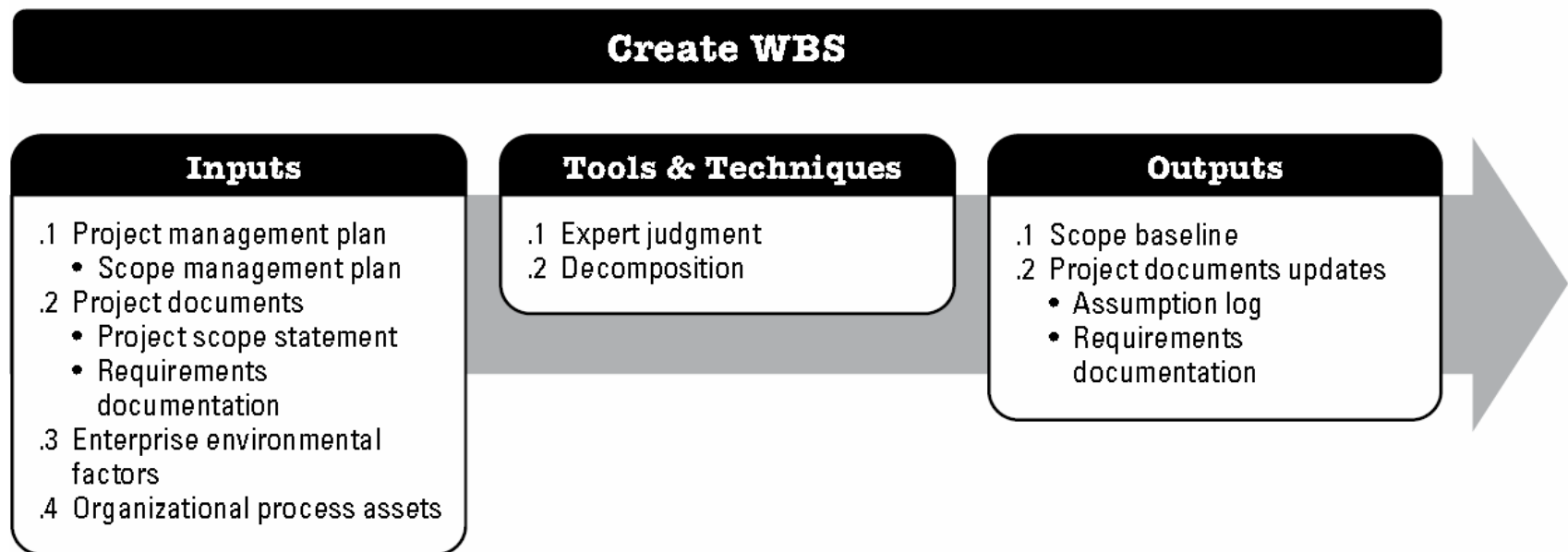
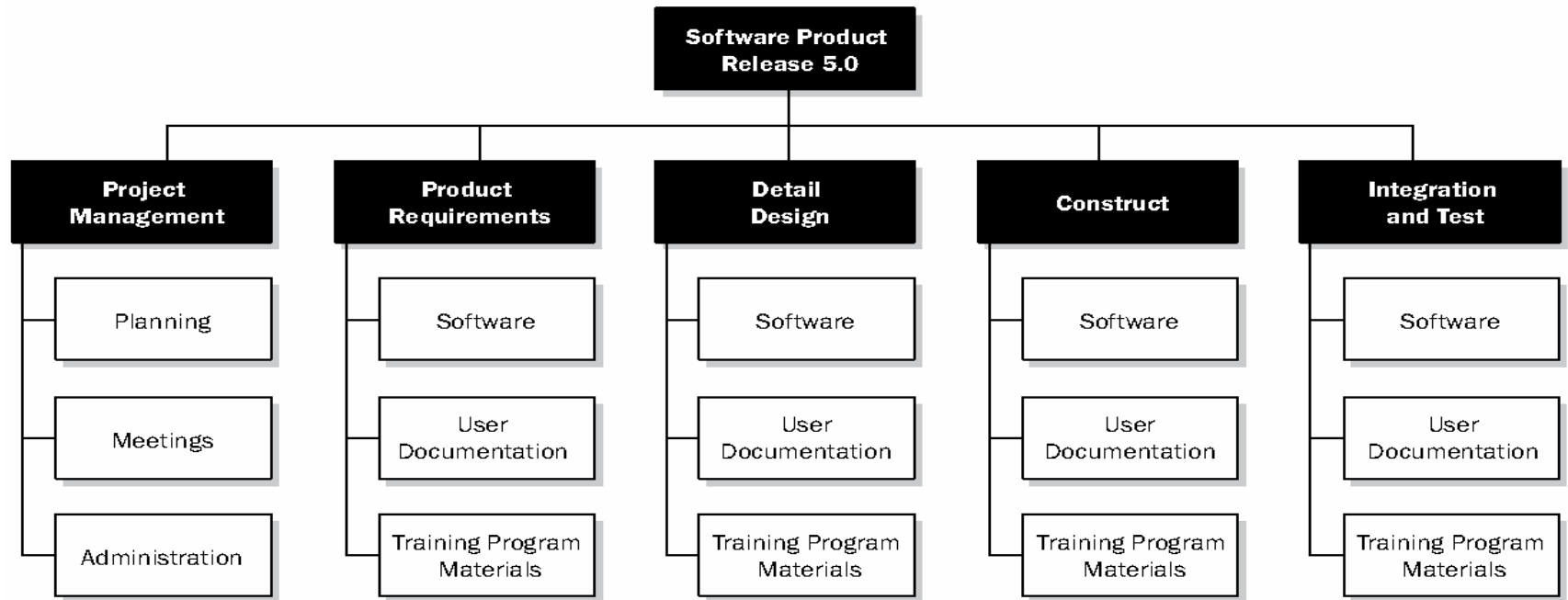


Figure 5-10. Create WBS: Inputs, Tools & Techniques, and Outputs

WBS – MECE Strategy

- ▶ MECE
 - ▶ Mutually Exclusive – No activity is overlapping
 - ▶ Collectively Exhaustive – The sum of small activity equal a big activity
- ▶ Level 0 – No activity
- ▶ Level 1 – Main deliverables
- ▶ Level 2 – End Deliverables
- ▶ Level 3 – Interim Deliverables

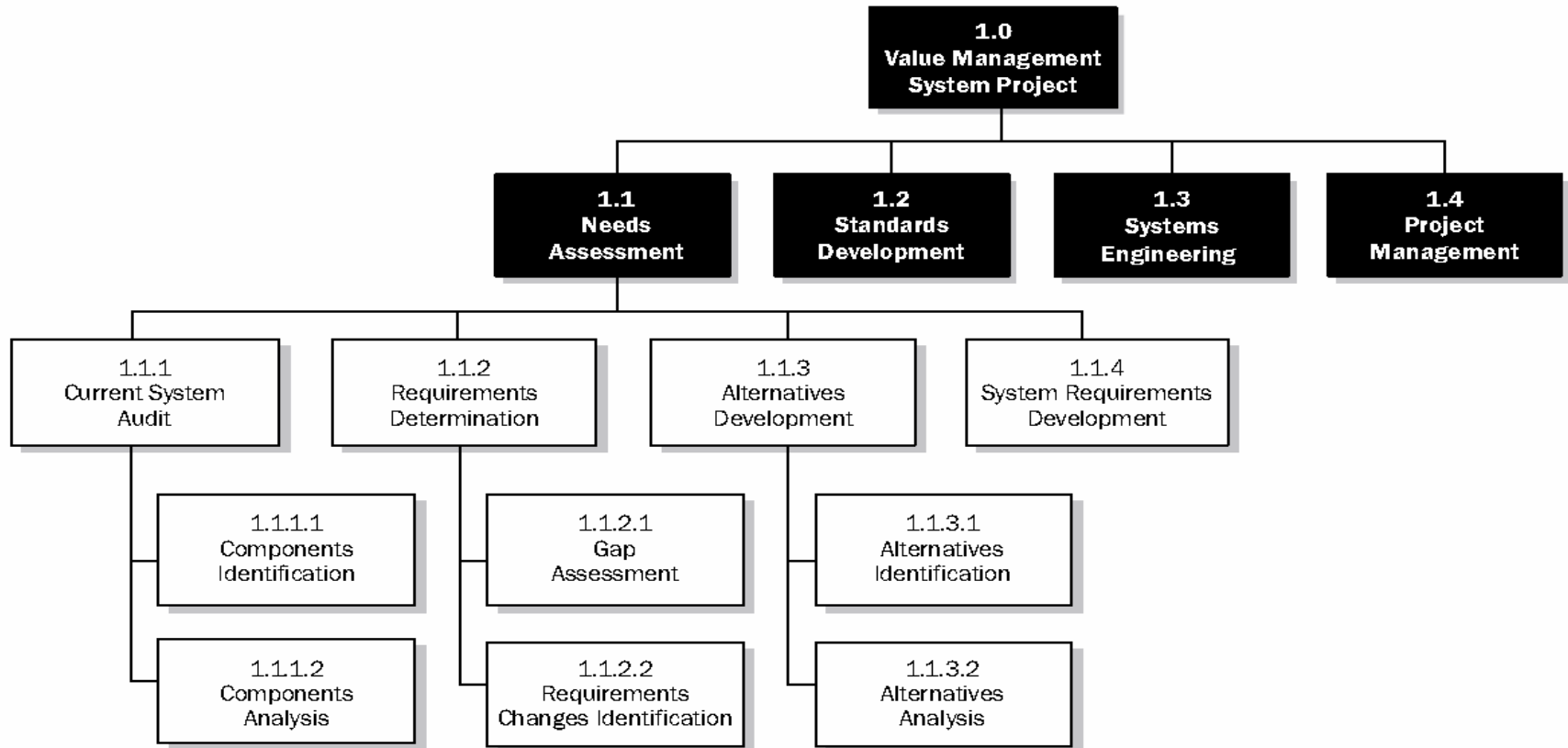
WBS – Decomposition by phase



The WBS is illustrative only. It is not intended to represent the full project scope of any specific project, nor to imply that this is the only way to organize a WBS on this type of project.

Figure 5-13. Sample WBS Organized by Phase

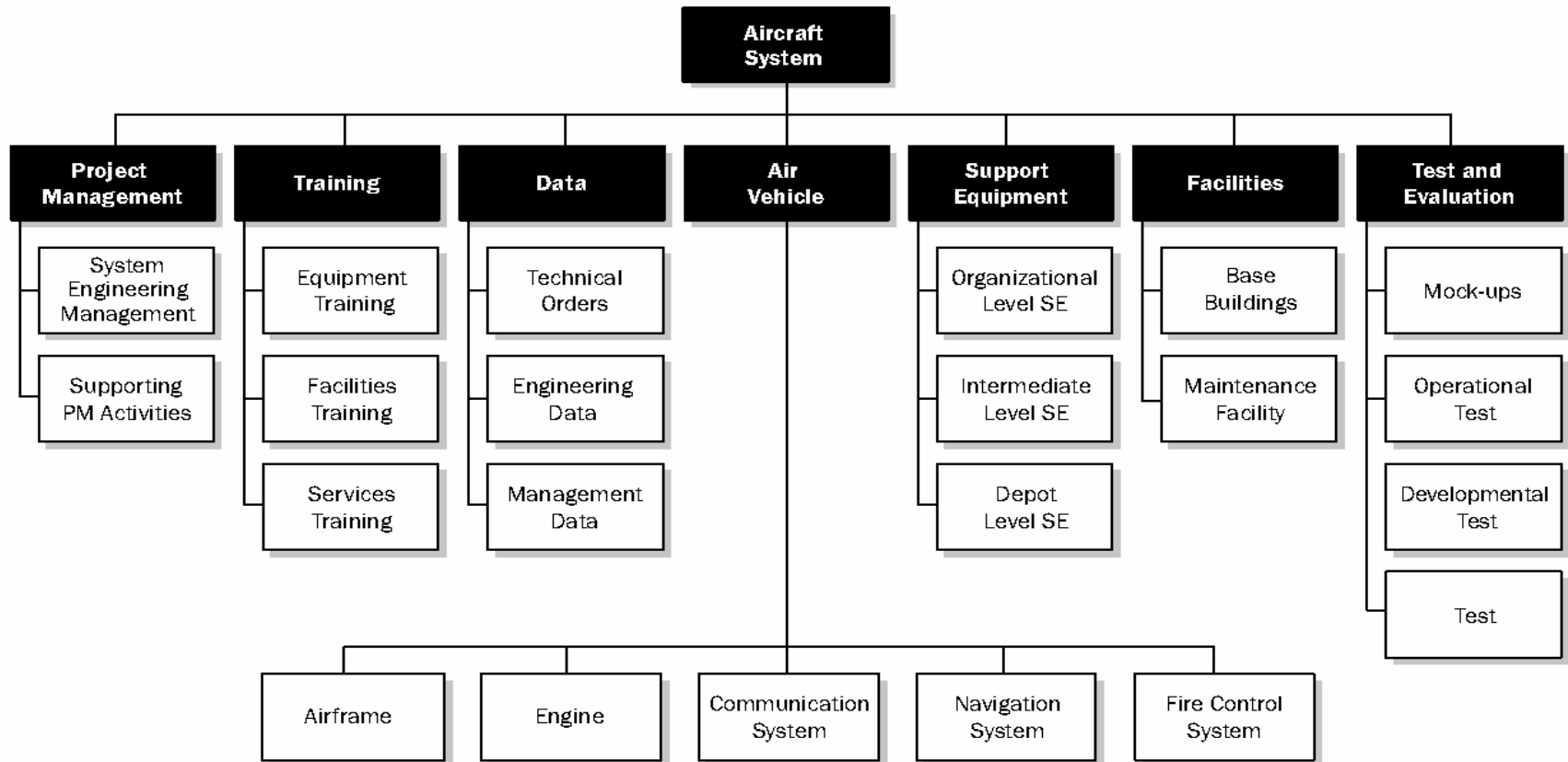
WBS – Decomposition by work package



The WBS is illustrative only. It is not intended to represent the full project scope of any specific project, nor to imply that this is the only way to organize a WBS on this type of project.

Figure 5-12. Sample WBS Decomposed Down Through Work Packages

WBS – Decomposition by phase



The WBS is illustrative only. It is not intended to represent the full project scope of any specific project, nor to imply that this is the only way to organize a WBS on this type of project.

Figure 5-14. Sample WBS with Major Deliverables

Create WBS – Scope Baseline

- ▶ Components of the scope baseline include:
 - ▶ **Project scope statement** --- the description of the project scope, major deliverables, assumptions, and constraints
 - ▶ **WBS** --- a hierarchical decomposition of the total scope of work to be carried out Each descending level of the WBS represents an increasingly detailed definition of the project work.
 - ▶ **Work package** --- The lowest level of the WBS is a work package with a unique identifier.

Validate Scope

- ▶ The key benefit of this process is that it brings objectivity to the acceptance process and increases the probability of final product, service, or result acceptance by validating each deliverable

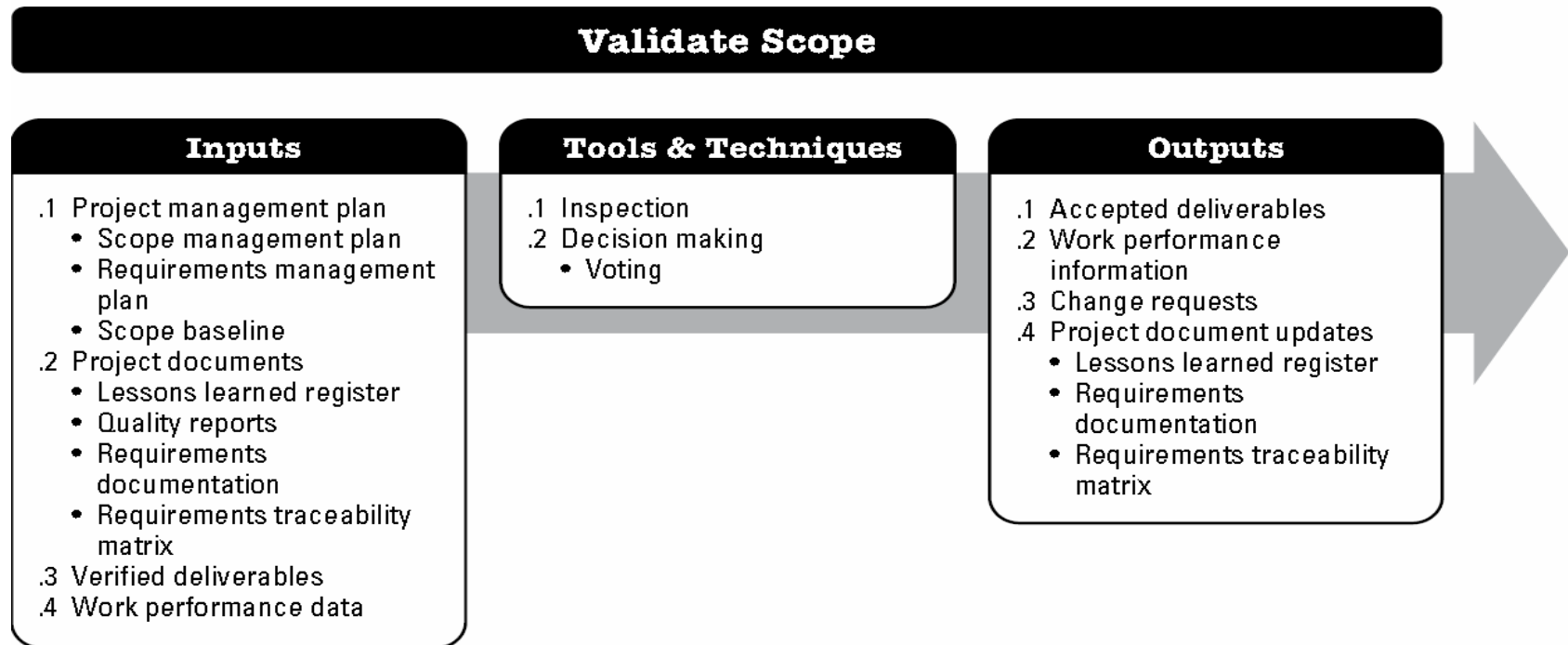


Figure 5-15. Validate Scope: Inputs, Tools & Techniques, and Outputs

Control Scope

- ▶ The key benefit of this process is that the scope baseline is maintained throughout the project.

